

Roll No.

TME-601**B. TECH. (ME) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022****REFRIGERATION AND AIR CONDITIONING****Time : Three Hours****Maximum Marks : 100**

- Note :** (i) All questions are compulsory.
(ii) Answer any **two** sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) Define the following terms : (CO1)
(i) Specific heat
(ii) Calculate the value of 1 TR
(iii) Air-Conditioning
(iv) Condensation
(v) Sensible heat
(b) Describe the Boot-strap Air Refrigeration system with neat schematic diagram. Also represent of boot-strap cooling system on T-s diagram. (CO1)

P. T. O.

- (c) A refrigerator working on Bell-Coleman cycle operates between pressure limits 1.05 bars and 8.5 bars. Air is drawn from the cold chamber at 10°C . Air coming out of compressor is cooled to 30°C before entering the expansion cylinder. Expansion and compression follow the law $PV^{1.35} = \text{constant}$. Determine C.O. P. of the system and refrigeration effect. (CO1)

Take the value of $\gamma = 1.4$ and $C_p = 1 \text{ kJ/kg-K}$ for air.

2. (a) Explain the working of simple Vapor Compression Refrigeration system (VCRs) with neat sketch diagram. (CO1, CO2)
- (b) A reversed Carnot cycle is used for cooling in summer maintained at 5°C when the ambient temperature is 40°C . The same cycle is used for heating in winter at 55°C when the ambient temperature is 15°C . Assuming 5°C temperature difference in the heat exchanger that exchanges heat with the surrounding, find : (CO1, CO2)
- (i) The COP for cooling and heating cycles.
- (ii) If the water at 30°C is used as cooling medium with 3°C temperature for summer air conditioning, what is will the Carnot C.O.P. ?
- (iii) The power consumption per ton of refrigeration in each case. Assume that the air and water temperatures remain constant during heat transfer.
- (c) An air refrigerator works between the pressure limits of 1 bar and 8 bar. The temperature of the air entering the compressor and expansion cylinder are 15°C and 35°C respectively. (CO1, CO2)

The expansion and compression follow the law $PV^{1.40} = \text{constant}$. Find the following :

- (i) Theoretical C.O.P. of the refrigerating cycle.
 - (ii) If the load on the refrigerating machine is 20 tons, then find the amount of air circulated per minute through the system, assuming the actual C.O.P. is 65% of theoretical C.O.P.
3. (a) What is refrigerant ? How is the refrigerant classified ? Give the chemical formula for R11, R22 and 134a. (CO2, CO3)
- (b) What are the advantages and disadvantages of using inorganic refrigerants ? Also write their names. (CO2, CO3)
- (c) What are the essential properties of a good refrigerant ? (CO2, CO3)
4. (a) Draw Li-Br absorption system showing all important components. State the functions of each component. (CO2, CO4)
- (b) Describe vapour absorption refrigeration system using three fluids. (CO2, CO4)
- (c) A refrigeration plant using CO_2 as refrigerant works between 25°C and -5°C . The dryness of CO_2 is 0.6 and compression end in the wet region. Find out C.O.P. of the system, assuming there is no sub-cooling of liquid refrigerant. Use the following table as properties of CO_2 . (CO2, CO4)

Take the following properties of CO_2 :

Temp. ($^\circ\text{C}$)	Liquid Heat (kJ/kg)		Liquid Entropy (kJ/kg-K)		Latent Heat (kJ/kg)
	Liquid	Vapour	Liquid	Vapour	
25°C	164.77	282.23	0.5978	0.9918	117.46
-5°C	72.57	321.33	245.7	1.2146	248.76

P. T. O.

5. (a) What are the different types of compressors used in refrigeration and air conditioning practice ? Discuss them. (CO5, CO6)
- (b) Explain the Psychrometry processes. Also represent them on psychrometry chart. (CO5, CO6)
- (c) Define the following terms : (CO5, CO6)
- Dew Point Temperature
 - Wet bulb Depression
 - Dew Point Depression
 - Specific Humidity
 - Relative Humidity

Temp. (°C)	Liquid Heat (kJ/kg)		Latent Heat (kJ/kg)	
	Liquid	Vapour	Liquid	Vapour
25°C	164.77	282.23	0.5978	0.9918
-5°C	100.7	321.33	242.7	1.2146

Roll No.

TME-602

**B. TECH. (ME) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

DESIGN OF MACHINE ELEMENTS-II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Design Data Handbook is permissible.

1. (a) Explain nipping in a leaf spring. List the materials commonly used for the manufacture of the leaf springs and write the expression for determining the stress and deflection in full length and graduated leaves.

(CO1)

(b) AS helical valve spring is to be designed for an operating load range of approximately 90 to 135 N. The deflection of the spring for the load range 7.5 mm. Assume a spring index of 10. Permissible shear stress for the material of the spring = 480 MPa and its modulus of rigidity = 80 kN/mm². Design the spring.

(CO1)

P. T. O.

- (c) Classify springs according to their shapes. Draw neat sketches indicating in each case whether stresses are induced by bending or by torsion. Explain the following terms of the spring : (CO1)

- (i) Free length
- (ii) Solid height
- (iii) Spring rate
- (iv) Active and inactive coils
- (v) Spring index
- (vi) Stress factor

2. (a) A V-belt drive consists of three V-belts in parallel on grooved pulleys of the same size. The angle of groove is 30° and the coefficient of friction 0.12. The cross-sectional area of each belt is 800 mm^2 and the permissible safe stress in the material is 3 MPa. Calculate the power that can be transmitted between two pulleys 400 mm in diameter rotating at 960 r.p.m. (CO2)

- (b) Give the application of the following wire ropes :

- (i) 6×7 rope
- (ii) 6×19 rope
- (iii) 6×37 rope

Under what circumstances a fibre rope and a wire rope is used ? What are the advantages of a wire rope over fibre rope ? (CO2)

- (c) An open belt 100 mm wide connects two pulleys mounted on parallel shafts with their centers 2.4 m apart. The diameter of the larger pulley is 450 mm and that of the smaller pulley 300 mm. The coefficient of friction between the belt and the pulley is 0.3 and the maximum stress in the belt is limited to 14 N/mm width. If the larger pulley rotates at 120 r.p.m., find the maximum power that can be transmitted. (CO2)

3. (a) A pair of spur gears consists of a 24 teeth pinion, rotating at 1000 r.p.m. and transmitting power to a 48 teeth gear. The module is 6 mm^2 , while the face width is 60 mm. Both gears are made of steel with an ultimate tensile strength of 450 N/mm. They are heat treated to a surface hardness of 250 BHN. Assume that velocity factor accounts for the dynamic load. Calculate (i) beam strength; (ii) wear strength; and (iii) the rated power that the gears can transmit, if service factor and the factor of safety are 1.5 and 2' respectively. (CO3)
- (b) A pair of parallel helical gears consists of an 18 teeth pinion meshing with a 45 teeth gear. 7.5 kW power at 2000 r.p.m. is supplied to the pinion through its shaft. The normal module is 6 mm, while the normal pressure angle is 20° . The helix angle is 23° . Determine the tangential, radial and axial components of the resultant tooth force between the meshing teeth. (CO3)
- (c) Write the expressions for static, limiting wear load and dynamic load for spur gears and explain the various terms used there in. Define formative or virtual number of teeth on a helical gear. (CO3)
4. (a) A pair of bevel gears consists of a 30 teeth pinion meshing with a 48 teeth gear. The gears are mounted on shafts, which are intersecting at right angles. The module at the large end of the tooth is 4 mm. Calculate (i) the pitch circle diameters of the pinion and the gear; (ii) the pitch angles for the pinion and gear; and (iii) the cone distance. (CO4)

- (b) A pair of worm gears is designated as 1/52/10/8. The worm rotates at 1000 r.p.m. and the normal pressure angle is 20° . Determine the coefficient of friction and the efficiency of the worm gears. (CO4)
- (c) A double threaded worm drive is required for power transmission between two shafts having their axes at right angles to each other. The worm has $14\frac{1}{2}^\circ$ involute teeth. The centre distance is approximately 200 mm. If the axial pitch of the worm is 30 mm and lead angle is 23° , find (i) lead; (ii) Pitch circle diameters of worm and worm gear; (iii) helix angle of the worm, and (iv) efficiency of the drive if the coefficient of friction is 0.05. (CO4)
5. (a) (i) Enumerate the applications of sliding-contact and rolling-contact bearings.
- (ii) Define L_{10} life, Dynamic bearing Capacity, Equivalent bearing load; Bearing Modulus, Sommerfeld's number. (CO5/CO6)
- (b) A ball bearing is subjected to a radial force of 2500 N and an axial force of 1000 N. The dynamic load carrying capacity of the bearing is 7350 N. The shaft is rotating at 720 r.p.m. Calculate the life of the bearing. (CO5/CO6)
- (c) A 360° hydrodynamic bearing operates under the following conditions :
Radial load 50 kN, Journal diameter = 150 mm, Bearing length = 150 mm, Radial clearance = 0.15 mm, Minimum film thickness = 0.03 mm, Viscosity of lubricant = 8 cP.
- What is the minimum speed of operation for the journal to work under hydrodynamic conditions ? (CO5/CO6)

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Roll No.

TME-603

B. TECH. (MECHANICAL ENGINEERING)
(SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
INTERNAL COMBUSTION ENGINES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the working of 2-stroke diesel engine with neat sketch. Differentiate between 2-stroke and 4-stroke diesel engines. (CO1)
- (b) In an engine working on dual cycle, the temperature and pressure at the beginning of the cycle are 85°C and 1.5 bar respectively. The compression ratio is 8. The maximum pressure is limited to 72 bar and total heat supplied per kg of air is 1865 kJ. Determine : (CO1)
 - (i) Pressure and temperatures at all salient points
 - (ii) Air standard efficiency
 - (iii) Mean effective pressure.

P. T. O.

- (c) In a constant volume "Otto cycle", the pressure at the end of compression is 15 times that at the start, the temperature of air at the beginning of compression is 38°C and maximum temperature attained in the cycle is 1950°C . Determine : (CO1)
- (i) Compression ratio
 - (ii) Thermal efficiency of the cycle
 - (iii) Work done
2. (a) Explain with the help of a P-V diagram the loss due to variation of specific heats in an Otto cycle. (CO2)
- (b) Discuss the important qualities of an SI and CI engine fuel. (CO2)
- (c) Compare the air-standard cycle and fuel-air cycles based on : (CO2)
- (i) character of the cycle
 - (ii) fuel-air ratio
 - (iii) chemical composition of the fuel
3. (a) Develop an expression for air-fuel ratio neglecting compressibility for a simple carburetor. (CO3)
- (b) Discuss the following with neat sketch : (CO3)
- (i) Individual pump and nozzle system
 - (ii) Unit injector system
- (c) Describe with suitable sketches the following system of a modern carburetor : (CO3)
- (i) Economizer system
 - (ii) Acceleration pump system
 - (iii) Choke

4. (a) Explain the four factors that influence the flame speed in detail. (CO4)
- (b) Explain the phenomenon of knock in CI engines and compare it with SI engine knock. (CO4)
- (c) Mention the effect of supercharging on engine performance. What are the limitations of supercharging in an IC engine? (CO4)
5. (a) The following observations were recorded in as test of one hour duration on a single cylinder oil engine working on 4-stroke cycle :

(CO5, CO6)

Bore = 280 mm; Stroke = 420 mm; Fuel used = 7.6 kg; Calorific value of fuel = 42810 kJ/kg; Average speed = 250 r.p.m.; map = 6.3 bar; Brake friction load = 1910 N; Quantity of cooling water = 710 kg; Temperature rise = 24°C, Diameter of brake wheel = 1.78 m.

Calculate :

- (i) Mechanical efficiency
- (ii) Brake thermal efficiency.

Draw heat balance sheet.

- (b) The following reading were taken during the test of a single cylinder four stroke oil engine :

P. T. O.

Cylinder diameter = 280 mm; Stroke length = 410 mm; Gross m. e. p. = 6.8 bar; Pumping m. e. p. = 0.3 bar; Engine speed = 280 r.p.m.; Net load on the brake = 1130 N; Effective diameter of the brake = 1.27 metres; Fuel used per hour = 12 kg; Calorific value of fuel = 43500 kJ/kg.

Calculate :

(CO5, CO6)

- (i) Indicated power
 - (ii) Brake power
 - (iii) Mechanical efficiency
 - (iv) Indicated thermal efficiency.
- (c) State the mechanism of formation of CO and HC. (CO5, CO6)

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TME-606

B. TECH. (SIXTH SEMESTER)

END SEMESTER EXAMINATION, 2022

CAD/CAM

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the applications of computers in design ? Write down the benefits of CAD. (CO1)

(b) What are the different graphic input and output devices used in CAD ? (CO1)

(c) Differentiate between the wireframe and solid modeling. (CO2)

2. (a) What are the different types of geometric models used in CAD ? (CO2)

P. T. O.

- (b) A triangle having vertices (1, 10), (5, 2), (8, 4) is translated by 2 units in Y-direction, then it is rotated by 45 degrees in counter-clockwise direction, then it is scaled by 3 units in X-direction. Find the final position of the triangle. (CO2)
- (c) For the points P_1 (1, 1), P_2 (2, 1), P_3 (4, 2), P_4 (2, 3) that define a 2D polygon develop a single concatenated transformation matrix that reflects about line $X = 0$. Translates by -1 in both X and Y-directions. Rotates by Z-axis by 180 degrees. Using this matrix, determine the new coordinates. (CO2)
3. (a) What is a part family? Write any five benefits of group technology. (CO3)
- (b) What are the material handling and control systems used in FMS? (CO3)
- (c) Explain the different types of FMS layout used in automation with the help of sketches. (CO3)
4. (a) What is automation? Why is it prime need now-a-days in most of the industries? (CO4)
- (b) What are the different NC methods used for motion or position control system? (CO4)
- (c) Explain briefly the open loop and the closed loop control system. (CO5)

(3)

5. (a) Differentiate between the manual and the computer aided part programming. (CO5)
- (b) Explain briefly the structure of the part program. Write down any *five* G and M codes used in the part programming. (CO6)
- (c) Write down the different components of the Robot. Briefly write its classification. (CO6)

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TME-614

**B. TECH. (ME) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

FINITE ELEMENT METHOD

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

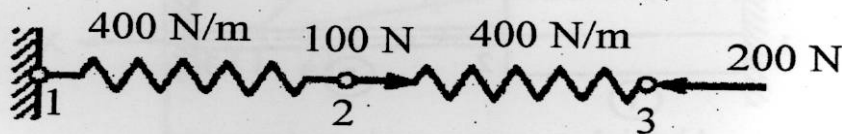
1. (a) (i) Describe the general steps of the finite element method.

(ii) Describe degrees of freedom, discretization and shape function.

(CO1, CO2)

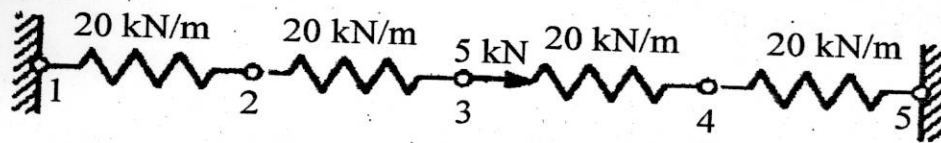
(b) For the spring assemblages shown in Figure, determine the nodal displacements, the forces in each element and the reactions. Use the direct stiffness method.

(CO1, CO2)

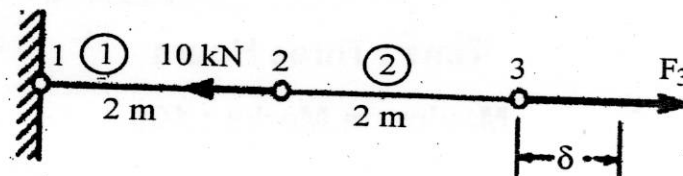


P. T. O.

- (c) For the spring assemblages shown in Figure, determine the nodal displacements, the forces in each element and the reactions. Use the direct stiffness method. (CO1, CO2)



2. (a) For the bar assemblages shown in Figure, determine the nodal displacements, the forces in each element and the reactions. Use the direct stiffness method. (CO2, CO3)

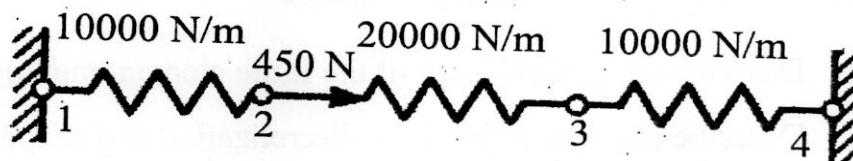


$$E = 210 \text{ GPa}$$

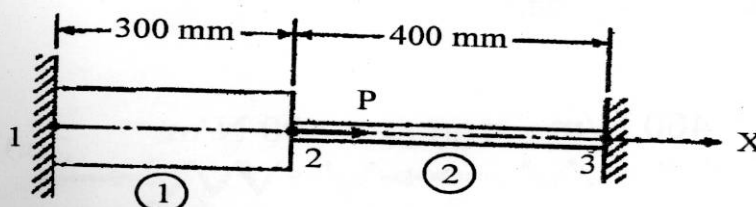
$$A = 4 \times 10^{-4} \text{ m}^2$$

$$\delta = 25 \text{ mm}$$

- (b) Use the principle of minimum potential energy to solve the spring problem shown in Figure below : (CO2, CO3)



- (c) Compute the nodal displacements, reaction forces for the bar assembly shown in Figure below : (CO2, CO3)



Aluminium

$$A_1 = 2400 \text{ mm}^2$$

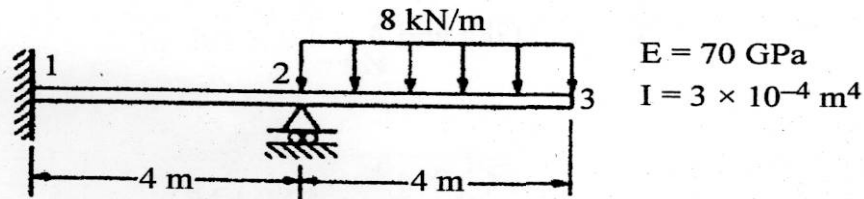
$$E_1 = 70 \times 10^9 \text{ N/m}^2$$

Steel

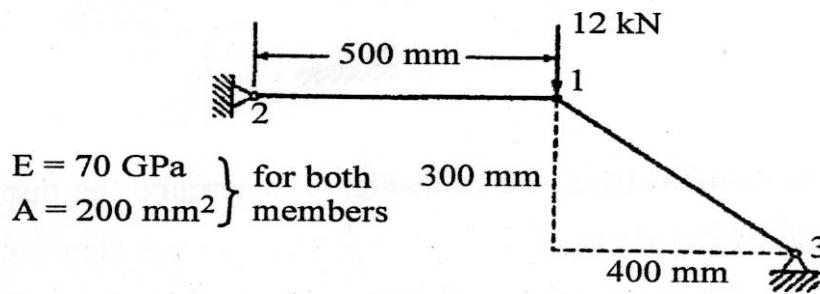
$$A_2 = 600 \text{ mm}^2$$

$$E_2 = 200 \times 10^9 \text{ N/m}^2$$

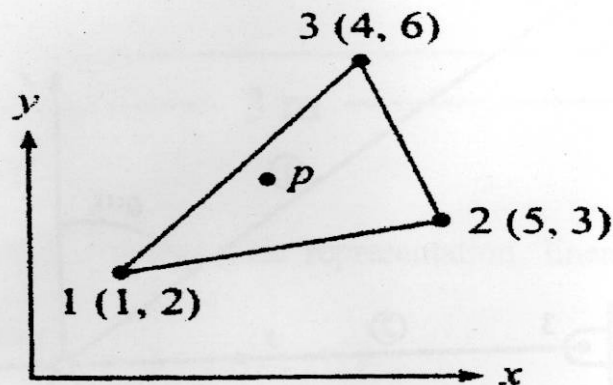
3. (a) For the beams shown in Figure, determine the nodal displacements and slopes, the forces in each element and the reactions. (CO5)



- (b) Compute nodal displacements for the truss elements shown in Figure below : (CO4)



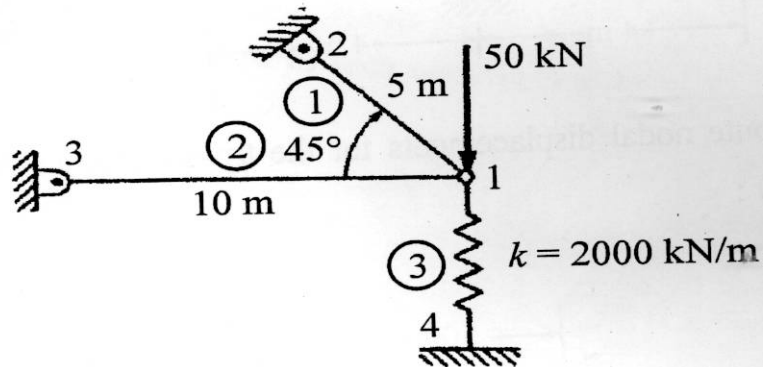
- (c) The nodal coordinate of triangular element is shown in Figure. At point P x-coordinate is 3.3 and $N_1 = 0.3$, determine N_2 and N_3 and y-coordinate at Point P. (CO4)



4. (a) For the plane trusses supported by the spring at node 1 show in Figure, determine the nodal displacements and the stresses in each element :

$$E = 210 \text{ GPa and } A = 4.0 \times 10^{-4} \text{ m}^2.$$

(CO4)

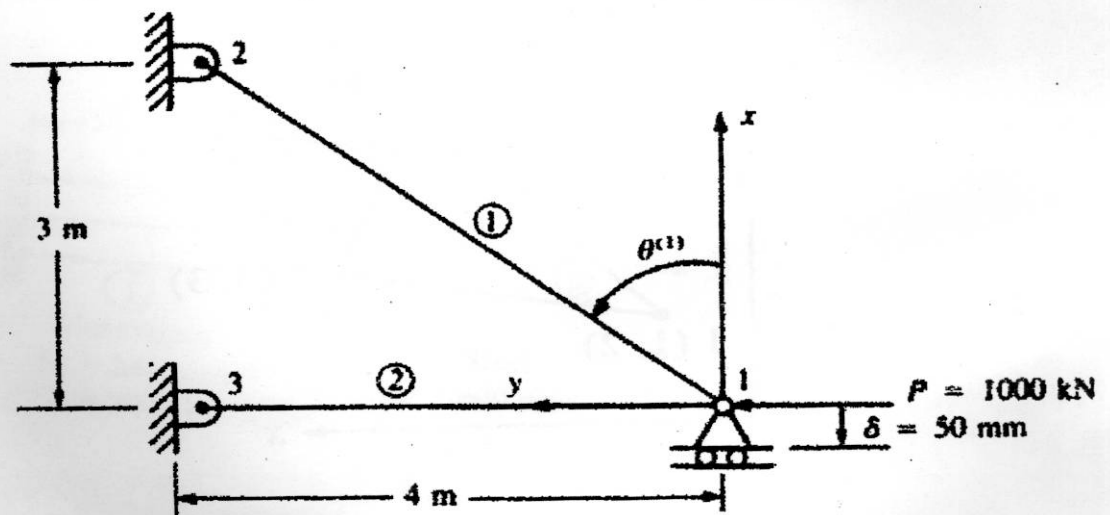


- (b) For the two-bar truss shown in Figure, determine the displacement in the y-direction of node 1 and the axial force in each element.

A force of $P = 1000 \text{ kN}$ is applied at node 1 in the positive y-direction while node 1 settles an amount $d = 50 \text{ mm}$ in the negative x-direction.

Let $E = 210 \text{ GPa}$ and $A = 6.00 \times 10^{-4} \text{ m}^2$ for each element. The lengths of the elements are shown in the Figure below :

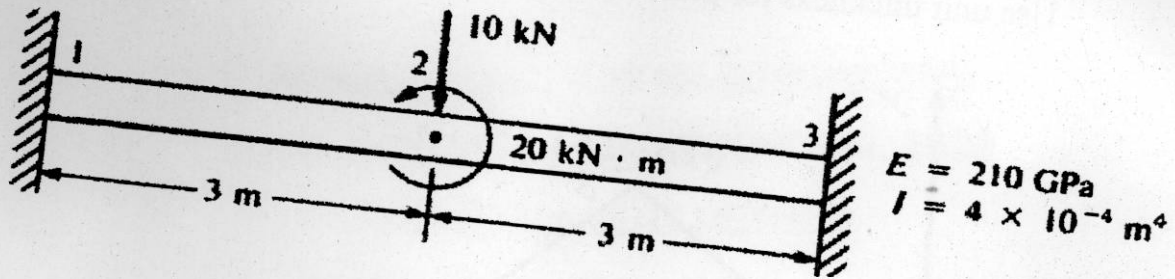
(CO4)



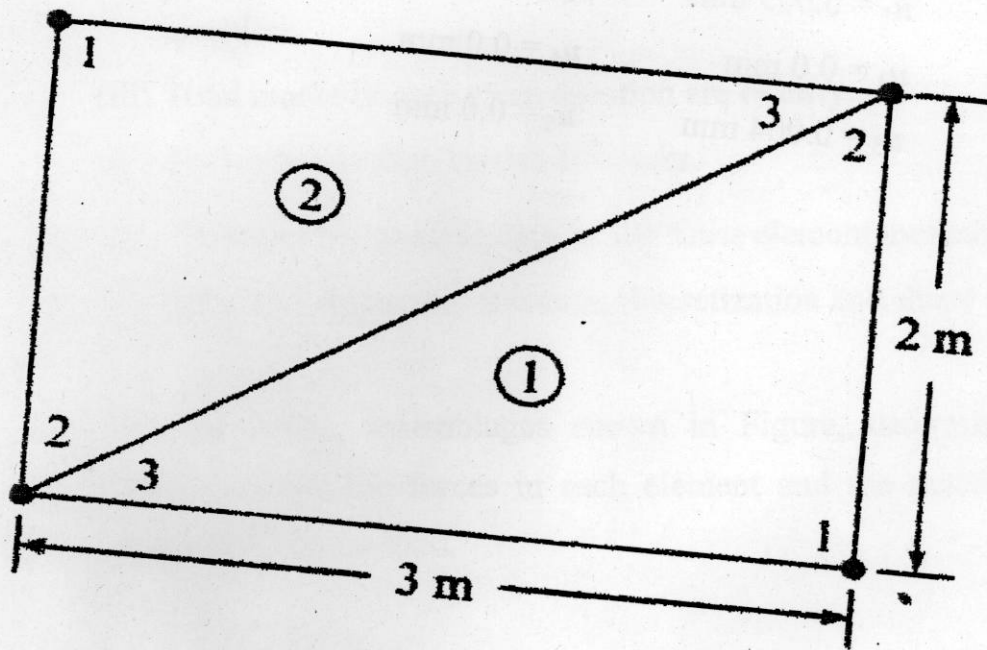
(5)

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- (c) For the beams shown in Figure, determine the nodal displacements and slopes, the forces in each element, and the reactions. (CO5)



5. (a) Determine the strain displacement matrices for the plate having two triangular elements. (CO6)



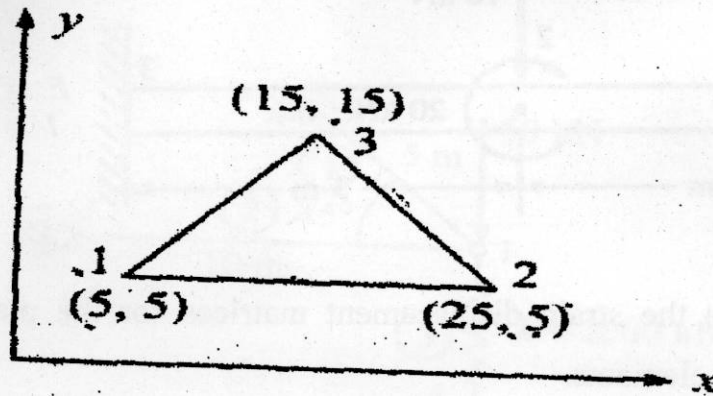
- (b) Explain in brief isoparametric representation, linear triangle element and CST element.

(CO6)

P. T. O.

- (c) For the plain element shown in Fig. the nodal displacements are given as follows. Determine the element stresses. Take $E = 200 \text{ GPa}$ $\nu = 0.3$. Use unit thickness for plain strain.

(CO6)



$$u_1 = 0.005 \text{ mm}$$

$$u_2 = 0.002 \text{ mm}$$

$$u_3 = 0.0 \text{ mm}$$

$$u_4 = 0.0 \text{ mm}$$

$$u_5 = 0.004 \text{ mm}$$

$$u_6 = 0.0 \text{ mm}$$

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Roll No.

TME-616

B. TECH. (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
NUMERICAL METHODS USING MATLAB

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) Using Taylor series estimate the approximation for second order derivative. (CO1, CO2)
- (b) Write a MATLAB code for calculating and plotting the sum of squares of first n natural numbers against n . (CO1, CO2)
- (c) Write a MATLAB code for declaring two 3×3 matrices, adding them and then calculating its inverse. (CO1, CO2)

P. T. O.

2. (a) Explain Gauss Seidel method in detail. (CO2, CO3)
- (b) Explain L and U matrices in LU decomposition. (CO2, CO3)
- (c) Solve for $x^2 - 4x - 7 = 0$ using Newton Raphson method. Show at least three iterations. (CO2, CO3)
3. (a) Calculate a_0 and a_1 for the following set of points for the linear regression (1, 1), (2, 3), (3, 3.5), (4, 3.7), (5, 6), (6, 5.3), (7, 7.9). (CO3, CO4)
- (b) How would you minimize the sum of square of errors while using polynomial regression ? (CO3, CO4)
- (c) When do you use multiple linear regression ? How do minimize the sum of square of errors ? (CO3, CO4)
4. (a) Why do you think the Simpson's rule can give you better result than trapezoidal rule ? (CO4, CO5)
- (b) Calculate the integral of $f(x) = x^2 + 4x - 2$ from $x = 0$ to 1 using trapezoidal rule. (CO4, CO5)
- (c) Calculate the integral of $f(x) = \sin(x)$ from $x = 0$ to $\pi/2$ using Simpson's 1/3 rule. (CO4, CO5)

(3)

5. (a) Using proper diagram, explain RK4 method. (CO5, CO6)
- (b) Can RK methods be used for solving boundary value problems ?
(CO5, CO6)
- (c) Formulate the differential equation $d^2y / dx^2 + p^2y = 0$ in the matrix eigenvalue problem using five internal points and finite difference method, $y = 0$ at $x = 0$ and L . (CO5, CO6)

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Roll No.

TME-615

**B. TECH. (ME) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

COMPUTATIONAL FLUID DYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Answers must be accompanied by neat and labelled diagrams.

(vi) Suitable assumptions can be made for any missing data.

1. (a) Discuss the limitations of the CFD approach. (CO1)
(b) Discuss the steps involved in solving a CFD problem with relevant example from Aerospace industry. (CO1)
(c) Describe how the continuity equation is discretized in the finite volume method. (CO2)
2. (a) Compare finite difference and finite volume methods. (CO2)

P. T. O.

- (b) Derive the following equation for a 2-dimensional case : (CO2)

$$a_p \phi_p = \sum a_{nb} \phi_{nb} + b$$

- (c) Solve the following system of equations using the Gauss-Seidel iterative method : (CO3)

$$2x + y + z = 5,$$

$$3x + 5y + 2z = 15$$

$$2x + y + 4z = 8$$

3. (a) Discuss about the different boundary conditions encountered in CFD problems along with relevant examples. (CO3)
- (b) Explain the SIMPLE method using the relevant equations. (CO3)
- (c) Compare implicit and explicit schemes with suitable examples. (CO3)
4. (a) State the RANS equations for turbulence flow and explain its importance. (CO4)
- (b) Explain Kolmogorov's theory and derive the different turbulence scales. (CO4)
- (c) Explain the concept of turbulent boundary layer. (CO5)
5. (a) What are the different types of Meshes in CFD ? Compare their advantages and disadvantages. (CO5)

(3)

(b) Distinguish between validation and verification of CFD solutions with a relevant example. What are the methods to reduce physical model uncertainty ? (CO6)

(c) Discuss about the guidelines for development and implementation of CFD solutions. (CO6)

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TAE-615

**B. TECH. (AME) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

VEHICLE TRANSPORT MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write down the need of Vehicle Transport Management. How Vehicle Transport Management is helpful in managing a fleet of transportation vehicles ? (CO1 and CO2)
- (b) Write down the different principles of transport management. Explain any *one* in detail. (CO1 and CO2)

P. T. O.

- (c) Which type of business ownership you prefer the most ? Explain in detail. (CO1 and CO2)
2. (a) Why scheduled maintenance is essential for a transportation vehicle ? How dose scheduled maintenance affect the overall profitability of the transport vehicle ? (CO2)
- (b) Classify different types of vehicle expenses. According to you which type of vehicle expenses mostly affects the profit margin. (CO2)
- (c) What is maintenance backlog ? How can it be managed ? (CO2)
3. (a) How will you manage cost against maintenance down time ? Explain in detail. (CO3)
- (b) How fleet management software are helpful in managing fleet activities ? Explain in appropriate words. (CO3)
- (c) What is inventory control ? Describe few inventory control methods. (CO3)
4. (a) Explain "Route Planning" and "Scheduling of Transport Vehicle" in appropriate words. (CO4 and CO5)
- (b) What are the different methods of fare collection ? Which one is most relevant in the context of current scenario ? (CO4 and CO5)
- (c) Prepare a route chart for pic-up and drop at customer's end using a suitable example of a well established business model. (CO4 and CO5)

(3)

5. (a) What are the different areas which have been covered under Motor Vehicle Act ? (CO6)

(b) What are the different categories of road sign in India ? Explain them briefly. (CO6)

(c) List down the different amendments in Indian Motor Vehicle Act.

(CO6)

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TAE-605

**B. TECH. (AME) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

IC ENGINE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the engine performance parameters. (CO1)
- (b) Explain the working of Diesel and Otto cycle with diagram. (CO1)
- (c) Discuss about the following : (CO1)
 - (i) Heat loss factor
 - (ii) Time loss factor
 - (iii) Exhaust blow down actor

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2. (a) Explain the flame propagation system in SI engines. (CO2)
- (b) What is the difference between the octane numbers and cetane numbers and list some of the advantages over each other. (CO2)
- (c) Describe the following terms : (CO2)
- (i) Fuel rating
 - (ii) Pre-ignition
 - (iii) Diesel knock (normal and abnormal combustion)
3. (a) Describe different methods of improving thermal efficiency of open cycle gas turbine plant. (CO3)
- (b) Explain the working and layout of gas turbine power plant with auxiliaries. (CO3)
- (c) Discuss about the merits and demerits of closed loop cycle turbine over open loop cycle turbine. (CO3)
4. (a) A single cylinder, 4 stroke cycle oil engines is fitted with a rope brake. The diameter of brake wheel is 600 mm and the rope diameter is 26 mm. The dead load on the brake is 200 N and the spring balance reads 30 N. If the engine runs 'it 450 r.p.m., what will be the brake power of engine ? (CO4, CO6)
- (b) Explain the heat balance sheet for SI engines. (CO4, CO6)
- (c) What is the importance of cooling system in IC engines and how does it hamper the engine performance ? (CO4, CO6)

(3)

5. (a) Explain the following terms :

(CO5, CO6)

(i) Fuel System

(ii) Ignition System

(iii) Starting System

(iv) Cooling System of IC engines.

(b) Explain the free piston engine and their types.

(CO5, CO6)

(c) Discuss about different components of transmission system.

(CO5, CO6)

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TRA-605

**B. TECH. (ME-RA) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

ROBOTICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the following in brief : (CO1)
robot robotics, fixed base robot, mobile robot, and degree of freedom.
(b) Describe the commonly used five robot arm configurations with neat sketches. (CO1)
(c) What are three laws of Robotics ? Also, give classification of robots according to JIRA. (CO1)
2. (a) A frame {Q} is rotated relative to a frame {P} about z-axis by 30 degree, translated 5 units in x, and translated 2 units in y; the z-axis is pointing out of the page. Position vector of a point with respect to frame

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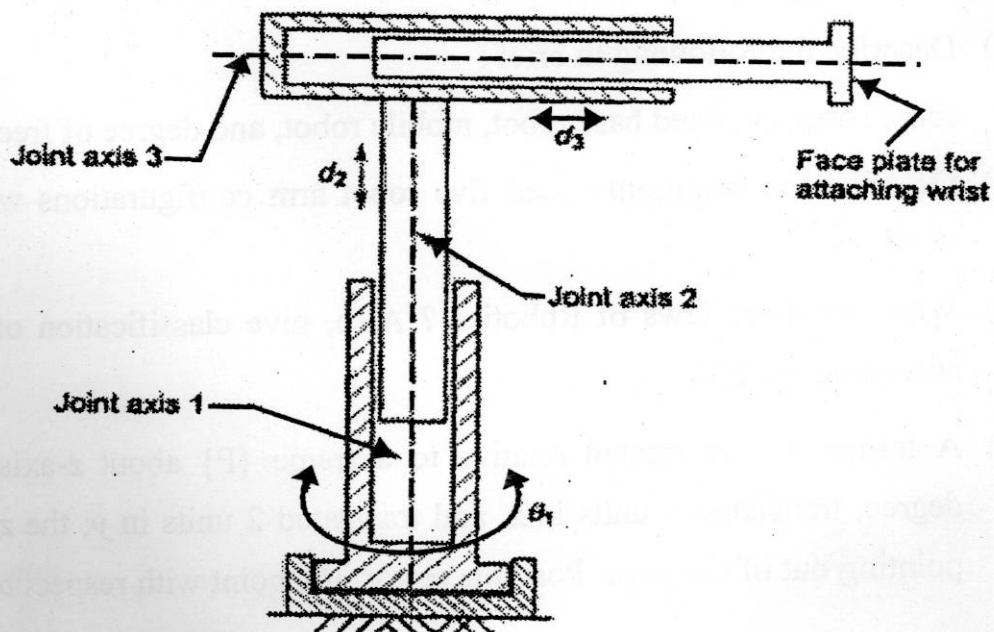
$\{Q\}$ is given by $[1 \ 3 \ 0]^T$. calculate position of that point with respect to frame $\{P\}$. (CO2)

(b) Describe the mapping in general frames with neat sketch and also discuss about the homogeneous transform. (CO2)

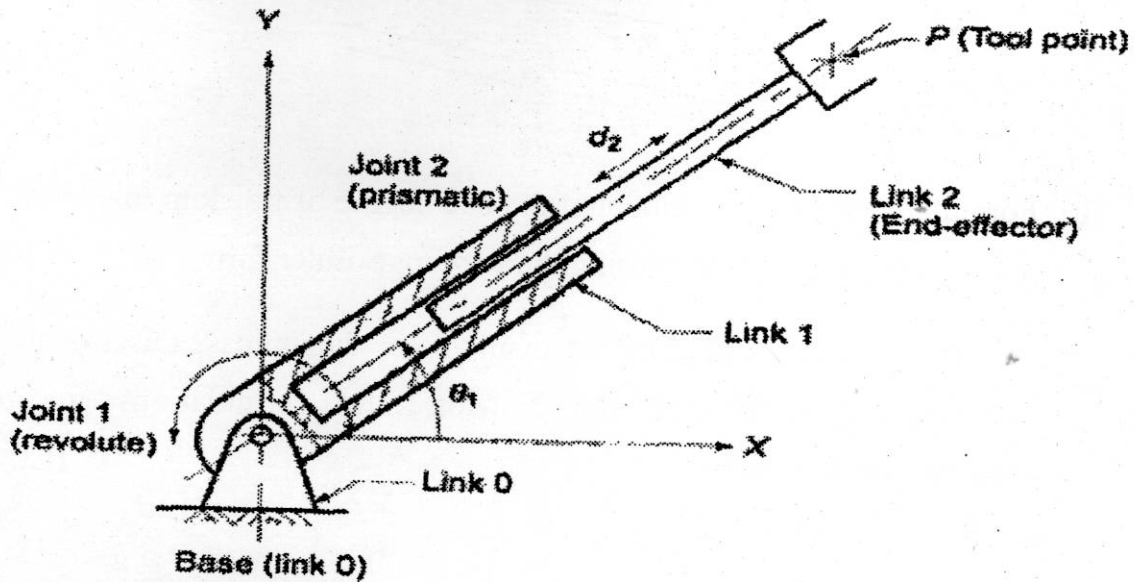
(c) A frame $\{B\}$ is rotated relative to another frame $\{A\}$ about z-axis by 45 degrees and translated 2 units in X and 1 unit in Y. The Z-axis is pointing out the page. Thus, we have a description of ${}^A_B T$. Find ${}^B_A T$. (CO2)

3. (a) What do you understand by the forward and inverse kinematics of manipulators ? Also, discuss about the D-H notation for robot kinematics. (CO3)

(b) Formulate the forward kinematic model for the 3-DoF, RPP manipulator shown in the figure below : (CO3)



- (c) Formulate the forward kinematic model for the 2-DoF, RP planar manipulator shown in the figure below : (CO3)



4. (a) What do you understand by the static force analysis of manipulators ? Also, discuss about the role of Jacobian matrix in the static force analysis. (CO4, CO5)
- (b) What do you understand by the manipulator differential motion ? Also, describe the relationship between Jacobian matrix, joint velocities vector and the end-effector velocities vector. (CO4, CO5)
- (c) Discuss about the following with mathematical relationship : (CO4, CO5)
- Jacobian inverse
 - Static force analysis

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TRA-614

B. TECH. (R & A) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022
MICROCONTROLLER AND EMBEDDED SYSTEM

Time : Three Hours**Maximum Marks : 100**

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Differentiate between microprocessor and microcontroller in detail. (CO1)
(b) Design half adder and full adder. (CO1)
(c) Discuss architecture of 8051 microcontroller with the help of a block diagram. (CO1)
 2. (a) Discuss various flags and register bank selection bits in Program Status Word register for 8051 microcontroller. (CO2, CO6)
(b) Explain any *ten* salient features of 8051 microcontroller. (CO2, CO6)
(c) Discuss immediate, direct and indirect addressing modes with the help of an example for each. (CO2, CO6)

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(2)

3. (a) Explain any *five* arithmetic instructions with the help of examples. (CO2, CO3)
- (b) Explain the instructions with the help of an example for each. (CO2, CO3)
- (c) Explain ROM organization of 8051 with the help of a diagram. (CO3, CO4)
4. (a) Write suitable instructions with proper format for the following operations :
- (i) Division
 - (ii) Increment
 - (iii) Rotate right
 - (iv) Exchange
 - (v) OR
- (b) Write short note on Arduino Uno. (CO3, CO4)
- (c) Write a program to add the values at memory locations 45H and 46H and store the results in locations 55H and 56H. (CO3, CO4)
5. (a) Differentiate between the following instructions : (CO3, CO5)
- (i) MOV and MOVX
 - (ii) INC and DEC
 - (iii) ADD and ADDC
 - (iv) MUL and DIV
 - (v) RL and RR
- (b) Derive look up table for a 7 segment LED indicating hexadecimal codes for display of digits from 0 to 9. (CO3, CO5)
- (c) Write a program to multiply the contents of location 30H and 31H. store result lower byte at 32H and higher byte at 33H. (CO3, CO5)

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TAS-605

**B. TECH. (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

AIRCRAFT PROPULSION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) Discuss different types of propulsion systems, describe it with suitable schematic and thermodynamics cycle to represent the whole process. Also, discuss how the thermal and overall efficiency can be increased ?

(CO1, CO2)

- (b) What are the aircraft engine design parameters ? Discuss them with suitable mathematical expressions ?

(CO2)

- (c) An internal combustion engine drives a propeller. The propeller has a diameter of 2.0 m and discharges air at a rate of 360 kg/s. The aircraft

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flight speed is 80.0 m/s. Determine the thrust, ideal, and actual power of the propeller, the Froude efficiency, and the pressure rise across the propeller. Assume air density to be 1.225 kg/m^3 . (CO3)

2. (a) You have to design a turbofan engine, which will be installed in an aircraft flying at 40,000 ft, and aircraft speed range is 0.75 to 0.85 Mach number.

The range of by-pass ratio is 4 to 6. Assume that the air passing through the core is in the range of 25-30 kg/s. Assume fuel-to-air ratio $f = 0.012-0.020$.

The other parameters are as follows : $P_{ec} = 60 \text{ kPa}$, $u_{ec} = 330 \text{ m/s}$, $A_{ec} = 0.36 \text{ m}^2$, $P_{eh} = 35 \text{ kPa}$, $u_{eh} = 540 \text{ m/s}$, $A_{eh} = 0.28 \text{ m}^2$. What will be the range of thrust and range of propulsive efficiency of the engine. Assume the suitable pressure and temperature at that height.

(CO3), (CO4)

- (b) Assume that you have to design a propulsive system for reconnaissance fighter plane. It has the following characteristics :

Flight Mach number : 0.88-0.95, Altitude 65,000 ft, Ambient temperature 216.7 K, Ambient pressure : 5.5 kPa, Fuel heating value : 45 MJ/kg, Thrust force range : 55-65 kN, Air mass flow rate : 35-45 kg/s, Assume the suitable Fuel mass flow rate, Aircraft gross weight (65,000 ft) : 170 kN, Aircraft take-off weight : 195 kN, Wing area :

25-35 m², Fuel weight : 57 kN, Maximum lift coefficient $CL_{max} = 1.9$, $CD_0 = 0.014$, $K1 = 0.22$, $K2 = 0.0$, Air density at take-off : 1.225 kg/m³, Air density at 650,000 ft : 0.88 kg/m³, Give the numerical value of the range of exit velocity of the gas from the engine, the range of thermal efficiency, the range of propulsive efficiency. (CO3)

- (c) Calculate the mass of the fuel consumed during a trip where its range is 4,000 km, flight speed is 250 m/s, $L/D = 10$, TSFC = 0.08 kg/N/h, and $m1 = 50,000$ kg in the following two cases : Head wind = 50 m/s, Tail wind = 50 m/s. (CO2)

3. (a) Give at least 3 ideas with schematic diagram for the integration of the turbojet and RAM jet engines. How these ideas can be incorporated for the flight which is flying at 100 km from the sea-level ? (CO5, CO6)

- (b) Describe the design steps of the turbojet, and turbofan engines with proper mathematical expressions and schematic diagram. (CO3)

- (c) An airplane is powered by four engines and has the following data :

(CO3, CO4)

Drag coefficient $Cd = 0.097$

Wing area $A = 260$ m²

Flight speed $u = 300$ m/s

Air density $\rho = 0.414$ kg/m³

Inlet area of each engine $Ai = 3.14$ m²

It is required to calculate the jet speed assuming that the nozzles are unchoked and the fuel-to-air ratio is negligible.

4. (a) Fuel consumption estimation is an important parameter in the design of an aircraft.

Assume lift-to-drag ratio 15-20. Fuel to air ratio 0.010-0.020. Assume suitable ratio between the weight of the aircraft at the end and start of cruise and take-off-mass of the aircraft. Design the following parameters : The range of the aircraft, the fuel consumed during the engine start, warming, and climb and fuel consumed in the whole trip. You can assume other parameters in suitable range. (CO4, CO5)

- (b) A Boeing aircraft have following data : Lift-to-drag ratio of 21. The fuel-to-air ratio is 0.012. Assume, the ratio between the weight of the aircraft at the end and start of cruise. Calculate the other parameters : The range of the aircraft. Calculate the fuel consumed during the engine start, warming, and climb and the fuel consumed in the whole trip. If the takeoff mass of aircraft is m_{MTO} kg ? Assume suitable parameters. (CO5)

- (c) Define and discuss the following with suitable expression : Thrust co-efficient, Torque co-efficient, Power co-efficient, Advance ratio, Reynolds number, Mach number. (CO1, CO2)

5. (a) Consider a RAM jet engine design to fly a vehicle at a Mach number of range 4-5 at an altitude of 60,000 ft. The engine has an inlet area of

$0.10 \text{ m}^2 - 0.20 \text{ m}^2$. The pressure ratio at combustion chamber is $P_{03}/P_{02} = 12-16$, and combustion efficiency is 92-96%. Assuming ideal diffuser ($P_{02} = P_{0a}$), At an altitude of 60,000 ft, the ambient temperature and pressure are 212.65 K and 15.75 kPa respectively. Calculate the air mass flow rate, maximum temperature, fuel-to-air ratio, exhaust velocity, thrust force. (CO5)

- (b) The flying machine powered by a pulsejet engine. It has the following data : The static thrust is 2500 N with an air inlet speed of 215 m/s. The intake area is 0.20 m^2 . The combustion efficiency is 0.92, The typical flight duration is 2000s, The exhaust temperature is 750 K. Assume that the specific heat of air is $C_{p_a} = 1.005 \text{ kJ/kg. K}$ and the specific heat of hot gases is $C_{p_h} = 1.12 \text{ kJ/kg. K}$. Calculate : The air-mass flow rate into engine, The exhaust velocity, The maximum temperature inside the engine. The maximum pressure, The fuel consumed during mission.

(CO5)

- (c) Draw a schematic diagram and discuss the working principle of the various types of rocket propulsion system. (CO2)